

Thin Origami Bellows

Lab Report

Bayu Febriansyah

677240334

April, 2026

AE 498 Space Structures Laboratory

University of Illinois Urbana-Champaign

Abstract

This study investigated the fabrication and quasi-static tensile deployment behavior of a thin origami bellows structure. The specimen was designed from a prescribed crease pattern with four repeating units ($N=4$), a facet width of 12 mm, and 45° crease angles, then modeled in Siemens NX, printed on paper, folded, and assembled into a tubular bellows. Mechanical testing was performed using an Instron universal testing machine under displacement-controlled axial tension, with the specimen mounted between fixture attachments and loaded from the compressed configuration. The measured response was analyzed using a Force vs. Deployment Percentage curve. The bellows exhibited a strongly nonlinear response, with the force increasing from an initial recorded value of 32.5 N to a maximum of 48.1 N at 13.58% deployment. Because the exported dataset began after a nonzero tensile load had already developed, the deployment values were evaluated using the recorded force range between the initial saved force and the maximum measured force. Based on this recorded force range, the deployment values corresponding to 10%, 20%, and 50% were 1.02%, 1.85%, and 5.03%, respectively. After the peak load, the force dropped abruptly, indicating structural failure rather than gradual softening. Visual observation confirmed that failure occurred by tearing in the crease region near the 3D-printed attachment. Overall, the results show that the bellows developed increasing resistance during early deployment, but its load-carrying capability was ultimately limited by localized failure near the fixture interface.

1. Introduction

Origami-inspired bellows are lightweight deployable structures capable of undergoing large, reversible shape changes through folding rather than through significant stretching of the material itself. Because they can compact into a small volume and later expand while maintaining an organized geometry, such structures are of interest in deployable engineering systems, especially in space applications where packaging efficiency and controlled deployment are important. In this lab, a thin paper origami bellows was fabricated and mechanically tested to examine how its folding geometry influences its tensile deployment response. The assigned bellows geometry consisted of four repeating units ($N=4$), a facet width of 12 mm, and 45° crease angles, as specified in the lab handout and crease-pattern drawing, Figure 1.

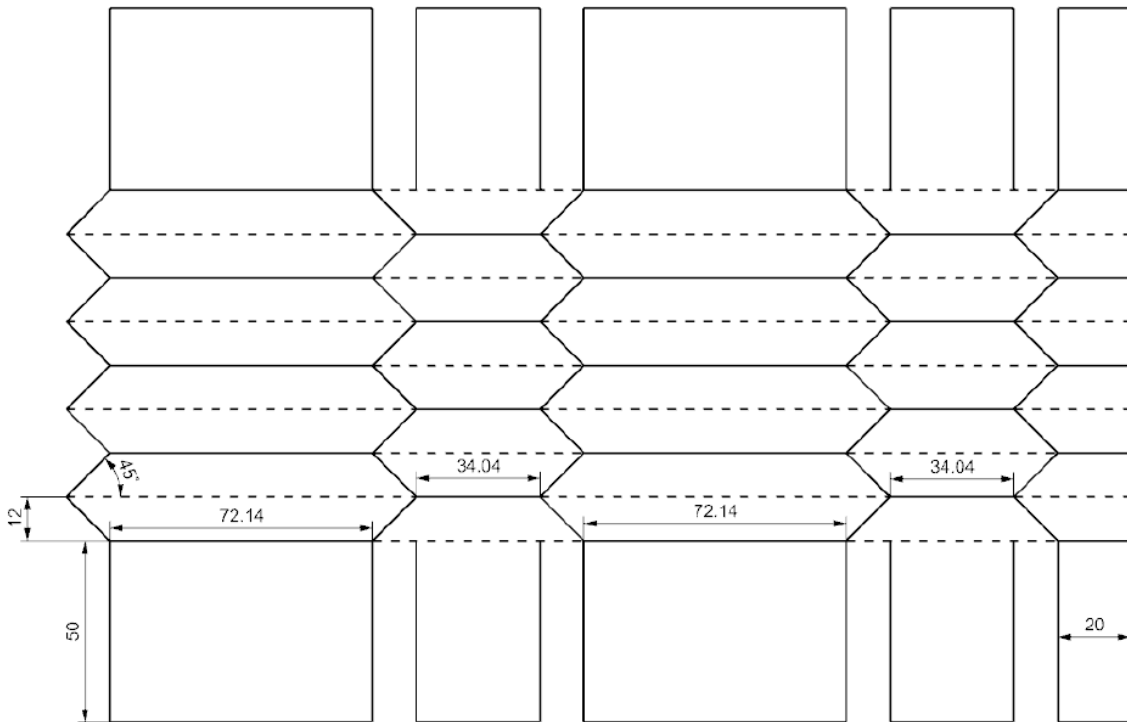


Figure 1. Reference crease pattern.

The first objective of the experiment was to translate the prescribed geometry into a CAD crease pattern and fabricate a working tubular bellows specimen. The lab procedure required the pattern to be modeled using the provided dimensions, printed on 11 in $\times$ 17 in paper, folded along the indicated mountain and valley creases, and joined along the end tabs using double sided tape to form a closed bellows that could compress and extend smoothly.

The second objective was to characterize the bellows under quasi-static uniaxial tensile loading using an Instron testing machine. The specimen was mounted from the fully compressed state, the force and displacement readings were set before testing, and tension was applied at a constant displacement rate until the bellows was fully deployed. The measured force and displacement data were then used to convert displacement into percent deployment, allowing the mechanical response to be presented in the required Force vs. % Deployment form.

Overall, the purpose of this experiment was to connect origami geometry, fabrication quality, and structural response. By comparing the physical bellows configuration with the measured deployment behavior, the experiment provides insight into how a folded thin-sheet structure resists opening, transitions through deployment, and carries load as its geometry changes. The resulting force-deployment curve therefore serves not only as a record of its mechanical response, but also as a signature of its unfolding behavior.

2. Methods

The bellows specimen was designed from the crease-pattern geometry prescribed in the lab handout. The assigned configuration consisted of four repeating units ($N=4$), a facet width of 12 mm, and 45° crease angles. The crease pattern was modeled in Siemens NX using the reference dimensions as shown in Figure 1, including the end tabs used to close the structure into a tube. The final CAD pattern was prepared for printing on 11 in $\times$ 17 in paper, Figure 2.

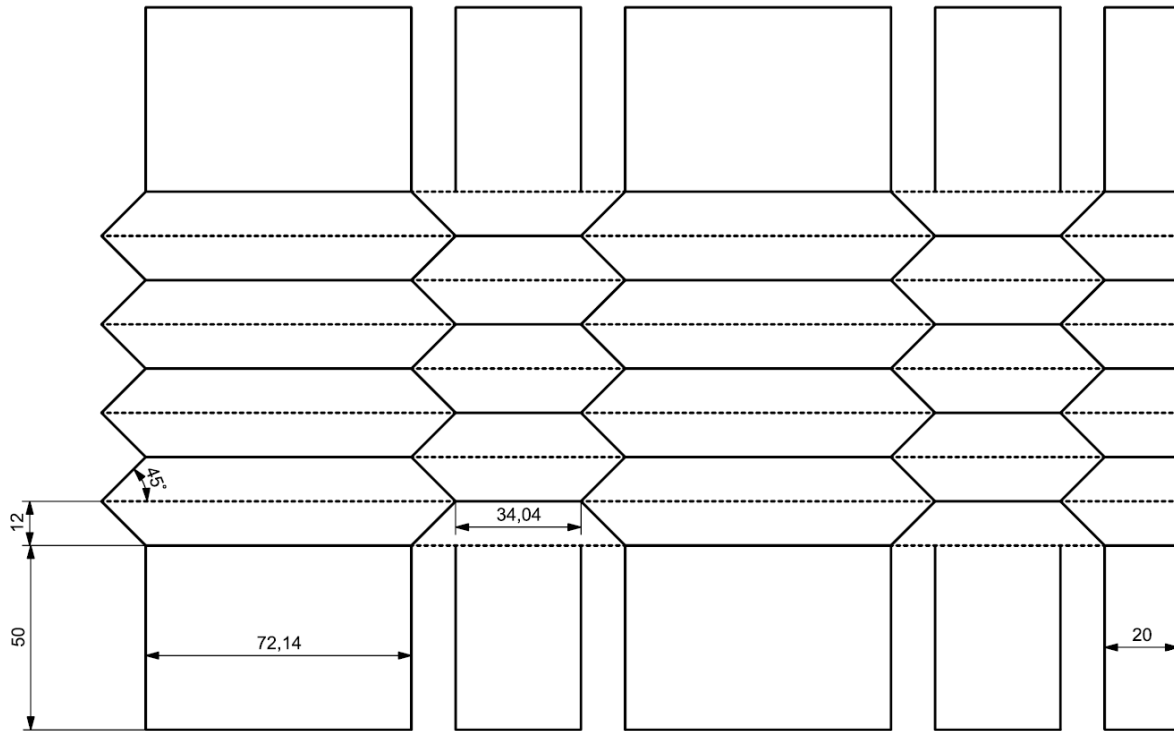


Figure 2. CAD crease pattern.

The CAD crease pattern was then printed and fabricated into a bellows by scoring and folding along the prescribed crease lines. The solid lines represent mountain folds and the dashed lines represent valley folds. The outer boundary was cut manually using scissors and cutter blades, after which the fold lines were scored and the pattern was folded into the required bellows geometry. The sheet edges were then joined to form a closed tubular specimen using double-sided Scotch tape along the seam. The paper type was not formally recorded, although it appeared to be slightly thicker than standard printer paper. This should be treated as an observational note rather than a precise material specification. After assembly, the bellows was checked qualitatively to confirm that it could compress and extend without obvious interference or visible damage, so that the specimen remained structurally sound during deployment. The finished specimen can be seen in Figure 3.



Figure 3. Mounted bellows specimen before testing.

Mechanical testing was performed on an Instron universal testing machine under quasi-static uniaxial tension. Through the lab handout and Nikhil's instructions, we established that the bellows should be mounted from the fully compressed state, the force and displacement should be zeroed before testing, and the specimen should then be loaded in tension at a constant displacement rate until fully deployed.

To mount the specimen, the bellows was first taped to a 3D-printed attachment, which was then secured to the machine using a clamp. The specimen was oriented vertically between the upper and lower fixtures. The bottom end remained fixed, while the top end moved upward with the Instron crosshead, producing axial tensile deployment of the bellows. Based on this configuration, the loading condition is best described as axial displacement-controlled tension with the specimen ends restrained through the fixture attachments. The ends were designed to behave approximately as clamped boundaries, although some compliance may have been introduced by the tape and attachment interface.

As a safety precaution, a mechanical stop/tab was positioned in the rolling mechanism of the Instron several centimeters above the estimated maximum displacement. This stop served as a travel limit to prevent the crosshead from continuing beyond the expected deployment range in the event of malfunction or other unexpected test behavior.



Figure 4. Instron test setup.

Figure 4 shows that the test was conducted on an Instron frame labeled 34TM-30, rated at 30 kN. No external extensometer was used, so the displacement recorded during testing was taken to be the crosshead displacement rather than a direct local measurement on the bellows. This is important because the measured displacement may include not only deployment of the bellows itself, but also small contributions from grip seating, attachment compliance, and tape deformation.

The loading rate was set to approximately 0.20 mm/s, or equivalent to 12 mm/min, to simulate quasi-static loading condition. The force and displacement data were sampled at approximately 20 Hz. Only one specimen was tested, and the reported dataset corresponds to a single trial. This should be acknowledged as a limitation when discussing repeatability, since no replicate tests were available for comparison.

After testing, deployment percentage was calculated using the formula below,

$$\text{Deployment Percentage} = \frac{D}{D_{max}} \times 100$$

where D is the displacement and D_{max} is the maximum recorded displacement. However, because the exported file did not begin at zero displacement, the raw machine displacement was first converted into a relative displacement using

$$\Delta D = D - D_0$$

where D_0 is the displacement at the first recorded point. Percent deployment was then evaluated using the relative displacement so that the test begins at 0% deployment and ends at 100% deployment at the maximum recorded extension. This adjustment is done so that the value is more physically consistent with the bellows starting from the compressed configuration.

The primary response used for analysis was therefore the Force vs. Deployment Percentage curve, constructed from the recorded axial force and the normalized relative crosshead displacement. Because the exported dataset began after a nonzero tensile load had already developed, the deployment values were evaluated using the recorded force range between the initial saved force and the maximum measured force, rather than directly from 10%, 20%, and 50% of the absolute maximum force. The exported dataset also contained machine-generated quantities such as stress, strain, and modulus. These values were not treated as the primary basis for interpretation because the specimen was a folded origami bellows rather than a uniform tensile coupon. As a result, the most meaningful outputs for this study were the axial force, crosshead displacement, and the derived percent deployment.

A key difficulty in the analysis was that the exported dataset began after measurable load had already developed. This was addressed by redefining displacement relative to the first recorded point and by evaluating the force-threshold quantities using the recorded force range.

3. Results

The bellows specimen was successfully loaded in quasi-static axial tension and produced a clear force response during deployment. Using the exported Instron dataset, the recorded crosshead displacement increased from 65.6994 mm to 70.1486 mm, corresponding to a total measured travel of 4.449 mm over 22.277 s. For analysis, this displacement was converted to relative displacement so that the first recorded point represented 0% deployment and the final recorded point represented 100% deployment, consistent with the deployment definition.

The mechanical response of the thin origami bellows under quasi-static axial tension is shown most clearly in the Force vs. Deployment Percentage plot, Figure 5. For the present analysis, the dataset was curated to the range from 0 to 20% deployment, which already captures the full load build-up, the peak response, and the subsequent loss of load-carrying capacity after the main failure event.

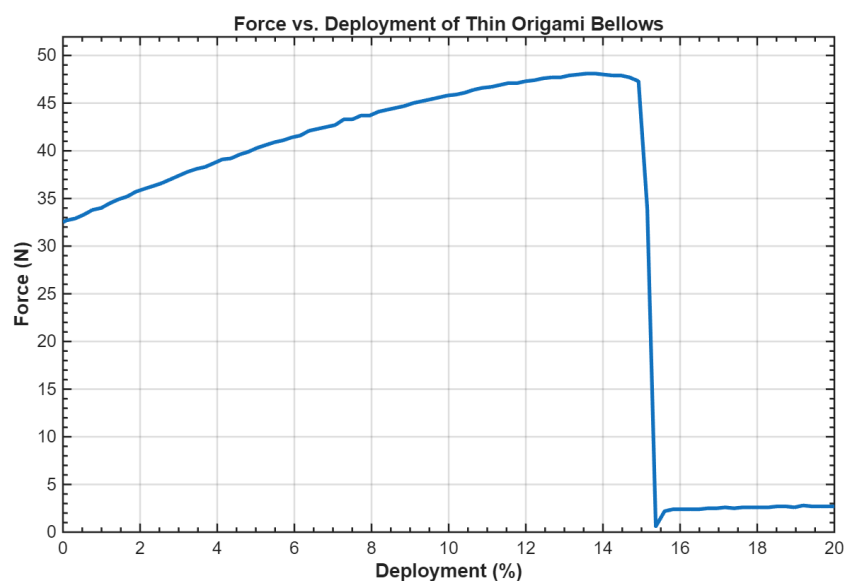


Figure 5. Force vs. Deployment Percentage over the analyzed 0–20% deployment range.

At the start of the recorded dataset, the bellows already carried an initial tensile force of 32.5 N at 0% recorded deployment. As deployment increased, the force rose gradually, indicating increasing resistance to opening of the folded geometry. The force reached a maximum value of 48.1 N at approximately 13.58% deployment. After this peak, the response remained near the maximum only briefly and then underwent a very sharp drop. Between approximately 15.15% and 15.37% deployment, the force fell abruptly from 33.9 N to 0.6 N, after which it remained at a low residual level of roughly 2.2–2.8 N through the end of the plotted range at 20% deployment.

This response indicates that the bellows initially resisted extension through geometric stiffness associated with the folded configuration, then reached a peak-load condition, and finally experienced a rapid loss of load-carrying capacity during deployment. In practical terms, most of the structural resistance was concentrated in the early portion of the extension history, while the post-peak regime was much more compliant. Because the force drop was very abrupt, the dominant mechanical event in the test was not gradual softening but rather a localized transition or failure-like release during deployment.

Next, the deployment values associated with 10%, 20%, and 50% of the recorded force range were determined. Because the exported dataset began after a nonzero tensile load had already developed, these values were evaluated using the force range between the initial recorded force and the maximum recorded force, rather than from a true zero-load baseline. Thus, we can obtain these values using the approach below.

$$F_{range} = F_{max} - F_{initial} = 48.1 \text{ N} - 32.5 \text{ N} = 15.6 \text{ N}$$

$$F_{10\%} = F_{initial} + 0.1(F_{range}) = 32.5 \text{ N} + 0.1(15.6 \text{ N}) = 34.06 \text{ N}$$

$$F_{20\%} = F_{initial} + 0.2(F_{range}) = 32.5 \text{ N} + 0.2(15.6 \text{ N}) = 35.62 \text{ N}$$

$$F_{50\%} = F_{initial} + 0.5(F_{range}) = 32.5 \text{ N} + 0.5(15.6 \text{ N}) = 40.30 \text{ N}$$

Using the Force vs. Deployment Percentage plot shown in Figure 5 and linear interpolation between neighboring data points, the corresponding deployment values can be obtained.

$$\text{Deployment at } F_{10\%} = 1.02\%$$

$$\text{Deployment at } F_{20\%} = 1.85\%$$

$$\text{Deployment at } F_{50\%} = 5.03\%$$

These results show that even relatively small amounts of early deployment produced a measurable increase in force, with half of the recorded force range already reached by about 5% deployment. The bellows continued stiffening up to the peak at about 13.6% deployment, after which the structure rapidly lost its ability to sustain load.

Visual observation during testing showed that the specimen failed by tearing in the crease region near the point where the bellows was attached to the 3D-printed attachment, which is consistent with the sudden force drop seen in the mechanical response.

4. Conclusion

This experiment successfully demonstrated the fabrication and tensile deployment behavior of a thin origami bellows with the prescribed geometry of $N=4$ repeating units, $W=12$ mm, and 45° crease angles. The bellows was designed from the assigned crease pattern, fabricated from folded paper, and tested in quasi-static uniaxial tension using an Instron machine, consistent with the objectives of the lab activity. The resulting force-deployment response showed that the structure developed increasing resistance during the early stages of extension, reached a clear peak load, and then underwent a sudden loss of load-carrying capacity.

The shape of the force-deployment curve provides the main mechanical insight from the test. During the initial portion of deployment, the force increased steadily, indicating that the folded bellows geometry offered increasing resistance to tensile opening. This behavior is consistent with a deployable structure in which the crease pattern and folded configuration govern the stiffness more strongly than the material alone. The bellows reached its maximum recorded force at relatively low deployment, showing that a large portion of its resistance was mobilized early in the extension process. After the peak, the curve dropped abruptly rather than decreasing gradually, indicating that the response was controlled by a failure event instead of smooth continued deployment of an intact structure. Since the lab specifically asks that the shape of the curve be discussed in relation to deployment behavior, the most important conclusion is that the specimen exhibited an initially stiffening response followed by a sudden failure-driven collapse.

Visual observation during testing confirmed that failure occurred by tearing in the crease region near the point where the specimen was attached to the 3D-printed fixture. This observation is important because it shows that the measured peak force should not be interpreted as a pure limit of the ideal bellows geometry alone. Instead, the maximum load was likely influenced by the local attachment condition, the taped interface, and stress concentration near the crease adjacent to the fixture. In other words, the experiment captured both the deployment mechanics of the origami bellows and the structural weakness introduced by the way the specimen was mounted.

Several limitations should be acknowledged. First, the analysis is based on one specimen and one trial, so no repeatability or specimen-to-specimen variation can be assessed. Second, the exported data began with a nonzero initial force, meaning the recorded response represents only the saved portion of the loading history rather than the full loading process from a true

zero-load state. Finally, because failure initiated near the attachment region, the measured peak response likely reflects both the bellows behavior and the influence of the fixture design. Because the exported dataset did not include the zero-load portion of the test, the deployment values at 10%, 20%, and 50% were evaluated using the recorded force range as a practical substitute.

Overall, the experiment met its main objective by showing how a thin origami bellows responds mechanically during tensile deployment. The results demonstrate that the structure can sustain increasing load during early unfolding, but that its usable deployment behavior is strongly limited by localized failure near the attachment. Future improvements would include testing multiple specimens, refining the attachment method to reduce stress concentration near the crease, and ensuring that data acquisition begins exactly at the zeroed state so that the full force-development history can be captured more accurately. These changes would make it easier to separate the intrinsic deployment mechanics of the bellows from artifact introduced by the boundary attachments and test setup.